

R for Earth and Sustainability Data Analytics

Instructor Resource Package

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Table of contents

Welcome	5
The R-GEO Analytical Ladder	5
How to use this manual	6
Why this book is easy to adopt	7
Student Success Stories	8
Instructor Language Guide	9
Sample Syllabus	10
Course description	10
Course-level learning outcomes	10
Prerequisites	10
Grading scheme	10
Academic integrity and responsible AI use	11
Calendar (Chapter 1 entry)	11
Chapter 1. Tool: Getting Started with R	12
Professor Cheat Sheet	12
Faculty Preparation Notes	13
Narrative, mindset and ladder position	17
A. Orient	19
B. Teach	21
C. Interpret	25
D. Practice	28
E. Research Extension	34
F. Career Pathways	37
G. Assess	39
H. Adapt and support	41
Chapter 2. Data: Data Handling and Preparation	43
Professor Cheat Sheet	43
Faculty Preparation Notes	44
Narrative, mindset and ladder position	49
A. Orient	50
B. Teach	54
C. Interpret	57
D. Practice	62
E. Research Extension	68
F. Career Pathways	70
G. Assess	72

H. Adapt and support	74
Chapter 3. Maps: Spatial Analysis (GIS)	77
Professor Cheat Sheet	77
Faculty Preparation Notes	78
Narrative, mindset and ladder position	83
A. Orient	84
B. Teach	88
C. Interpret	92
D. Practice	97
E. Research Extension	102
F. Career Pathways	105
G. Assess	107
H. Adapt and support	109
Chapter 4. Time: Time Series Fundamentals	112
Professor Cheat Sheet	112
Faculty Preparation Notes	113
Narrative, mindset and ladder position	118
A. Orient	119
B. Teach	122
C. Interpret	126
D. Practice	130
E. Research Extension	136
F. Career Pathways	139
G. Assess	141
H. Adapt and support	143
Chapter 5. Models: Modeling, Prediction, and Machine Learning	146
Professor Cheat Sheet	146
Faculty Preparation Notes	147
Narrative, mindset and ladder position	152
A. Orient	153
B. Teach	156
C. Interpret	161
D. Practice	165
E. Research Extension	171
F. Career Pathways	174
G. Assess	176
H. Adapt and support	178
Chapter 6. Earth Systems: Earth System Case Studies	181
Professor Cheat Sheet	181
Faculty Preparation Notes	182
Narrative, mindset and ladder position	188
A. Orient	189
B. Teach	192
C. Interpret	196

D. Practice	201
E. Research Extension	207
F. Career Pathways	210
G. Assess	211
H. Adapt and support	214
Chapter 7. Sustainability: Global Change and Sustainability	217
Professor Cheat Sheet	217
Faculty Preparation Notes	219
Narrative, mindset and ladder position	224
A. Orient	225
B. Teach	228
C. Interpret	232
D. Practice	237
E. Research Extension	243
F. Career Pathways	247
G. Assess	248
H. Adapt and support	251
Assessment Bank	253
Quiz 1: Chapter 1 (Tool)	253
The Three Master Rubrics	256
Analysis rubric	256
Communication rubric	256
Research rubric	257
Appendix: Answer Keys	258
Lab 1 (Getting started): model answers	258
Quiz 1: answer key	259

Welcome

This Instructor Resource Package helps you teach an introductory **Earth Data Analytics with R** course directly from *R for Earth and Sustainability Data Analytics*. It is built on one idea from the book itself: “the code is only the means; scientific interpretation is the goal.” You are not teaching a programming course. You are helping students learn to think like Earth Data Analysts who use R as a tool.

Everything here is designed to lower your preparation time to nearly zero, and to work even if you are an expert in Earth or sustainability topics but not a confident R programmer.

The R-GEO Analytical Ladder

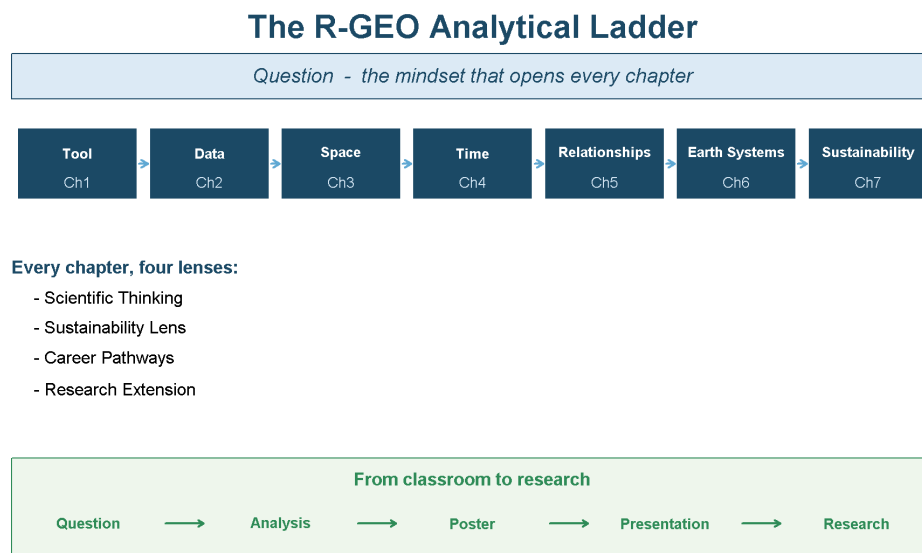


Figure 1: The R-GEO Analytical Ladder: the seven chapters as rungs, the four per-chapter lenses, and the path from a question to research.

The course is organized as a ladder. **Question** is the mindset that opens every chapter: every analysis starts with a scientific question worth answering. The seven climbable rungs then map one to one onto the seven book chapters.

Rung	Chapter	Students learn to
Tool	Ch1	use R, and think computationally, as the tool that makes the climb possible
Data	Ch2	prepare trustworthy data and judge what it can support
Space	Ch3	understand spatial structure
Time	Ch4	understand temporal structure
Relationships	Ch5	model relationships, predict, and meet machine learning
Earth Systems	Ch6	apply analytics to real Earth and sustainability systems
Sustainability	Ch7	integrate spatial and temporal global change and sustainability

Each chapter also carries four lenses: scientific thinking, a sustainability lens, career pathways, and a research extension.

How to use this manual

Every chapter module follows the same structure, so you always know where to look.

Part	What it gives you
Professor Cheat Sheet	One page that stands alone: ladder position, learning goals, key ideas, common mistakes, one activity, one assessment idea. If you read nothing else, read this.
Faculty Preparation Notes	Preparation time, what to review beforehand, likely student questions, troubleshooting, what can be safely skipped, and a minimum viable lesson plan. Written for instructors who are not R experts.
Narrative and Sustainability Lens	Where students are on the ladder, the driving question, and how the chapter connects to sustainability.
Orient, Teach, Interpret	Objectives, misconceptions, lecture outline, analogies, active-learning activities, and how to interpret each figure scientifically.
Practice	Coding exercises, project ideas, and AI activities that build thinking rather than code generation.
Research Extension and Career Pathways	How chapter work becomes a poster or research project, and which careers it connects to.

Part	What it gives you
Assess and Adapt	Tiered exam questions, the master rubric to apply, and modifications for different levels and for online or flipped teaching.

Grading uses only three master rubrics (Analysis, Communication, Research), defined once in this manual. Chapter modules tell you which rubric to apply; they never invent new ones.

Why this book is easy to adopt

- **Course-ready, near-zero preparation.** Slides, labs, assessments, and a sample syllabus are provided.
- **Teachable even if you are not an R expert.** Every chapter has Faculty Preparation Notes.
- **No prerequisites for students.** No prior coding, GIS, statistics, or Earth-science background is assumed.
- **Teaches scientific thinking, not syntax.** The Ladder keeps the focus on Earth Data Analytics.
- **Relevance in every chapter.** A sustainability lens, real career pathways, and a path from classroom work to undergraduate research.

Student Success Stories

One of the strongest reasons to adopt this book is simple: beginners succeed with it. Students who arrive with no coding, GIS, statistics, or Earth-science background have gone on to analyze real data, build posters, and present at conferences.

The cases below show the path from a starting point to a real outcome. They are not testimonials. Each one names a starting point, a project, and an outcome, so a prospective instructor can see what is realistic for their own students.

! Important

Replace every placeholder below with a real, verifiable case before publication. Do not invent students, projects, or outcomes.

Learner	Starting point	Project	Outcome
High-school student	[author to supply real case]	[author to supply real case]	[author to supply real case]
Undergraduate student	[author to supply real case]	[author to supply real case]	[author to supply real case]
Conference poster	[author to supply real case]	[author to supply real case]	[author to supply real case]

When you fill these in, keep each cell to one short phrase. The goal is for a professor to read the row in a few seconds and think: “a student like mine could do this.”

Instructor Language Guide

The words we use shape whether a beginner feels capable or discouraged. Many instructors discourage students without meaning to. This one-page guide keeps classroom language consistent with the encouraging tone of the book. The pattern is always the same: move attention away from “you should already know this” and toward “the idea is what matters, and the skill grows with practice.”

Avoid saying	Why it hurts	Prefer saying
“This is easy.”	If the student finds it hard, they conclude something is wrong with them.	“Let us take this one step at a time.”
“Everyone should already know this.”	It shames the very beginners the book is written for.	“This is new for many people, and that is fine.”
“Just memorize this function.”	It signals that the course is about syntax, not thinking.	“Focus on the idea. The syntax becomes familiar with use.”
“That is a silly mistake.”	It makes students afraid to run code and make errors.	“Errors are part of learning to code. Let us read what it is telling us.”
“The answer is obviously X.”	“Obvious” hides the reasoning the student needs.	“Here is how we can reason our way to the answer.”
“Did you get the right number?”	It rewards a result over understanding.	“Does this result make scientific sense?”

A useful habit: before correcting a student, name the idea they were reaching for. Beginners struggle with abstraction, not with effort. When concepts become hands-on and the language stays kind, they become meaningful.

Sample Syllabus

Note

This pilot shows the course framing and the Week 1 to 2 (Chapter 1) entry. The full 15-week calendar and the alternate tracks (compressed or summer, high school, and business school) are completed in a later phase.

Course description

An introduction to Earth Data Analytics using R. Students learn to ask scientific questions about Earth and sustainability systems, prepare and analyze real data, and interpret results responsibly. R is taught as a tool, not as the subject. No prior coding, GIS, statistics, or Earth-science background is required.

Course-level learning outcomes

By the end of the course, students will be able to:

- ask a focused scientific question about an Earth or sustainability system and choose data that can address it,
- read and run R scripts, and recognize common patterns, without memorizing commands,
- visualize spatial and time-based patterns and judge what they do and do not show,
- build and interpret simple models, and state their limits,
- and approach a new dataset with a structured, reproducible workflow.

Prerequisites

None. The course is designed for true beginners.

Grading scheme

Grading uses the three master rubrics defined in this manual. A suggested weighting:

Component	Weight	Rubric
Weekly labs	30%	Analysis
Two projects	30%	Research
Midterm and final	30%	Analysis and Communication
Participation and discussion	10%	Communication

Academic integrity and responsible AI use

AI tools are allowed as a learning accelerator, not as a replacement for thinking. Students may use AI to explain concepts, debug, and check interpretations, but must be able to explain their own work. See the AI in Education framework in this manual for sample policy language.

Calendar (Chapter 1 entry)

Week	Ladder rung	Chapter	Focus	Deliverable
1 to 2	Tool	Ch1	Getting started with R and RStudio: objects, vectors, functions, data frames, packages, a first map, a first plot. Emphasis on reading and interpreting code, not memorizing it.	Lab 1 (Analysis rubric)

Chapter 1. Tool: Getting Started with R

Professor Cheat Sheet

Field	Content
Ladder position	Tool (Chapter 1). The first rung. Students arrive with no coding background. They leave able to run an analysis and read a result, so the rest of the ladder is reachable. Driving question: “How do I even run an analysis?”
Sustainability Lens	Before you can ask whether an aquifer is drying or emissions are rising, you need a tool that reads Earth data the same way every time. Reproducibility is a sustainability skill: a result others can re-run is a result a community can act on.
Conference output	A student who can load <code>maps</code> , plot two vectors, and read a <code>volcano</code> Day 1 / Day 2 temperature map already has the mechanical skills behind a first poster figure. The reproducible script is the methods section. (<code>volcano</code> is R’s built-in elevation grid of a real hill, which we relabel as temperature for teaching.)
Learning goals	Treat R as one interchangeable tool for reading Earth data, not as the subject. Read code from the inside out and predict what one line produces before running it. Recognize that data structures (vector, data frame, array) mirror how real Earth data are shaped (a record, a table, a stack of maps).
Key ideas	An object stores a value so you reuse it, not retype it. A vector is one measured record; a data frame is a station table; an array is a stack of maps over time. A function takes input and returns output; read nested calls inside-out. Packages (<code>maps</code> , <code>leaflet</code>) add tools; install once, load each session. A plot turns numbers into a pattern you can question.
Common mistakes	Running code only in the Console, then losing it. Mixing text and numbers in one vector (the “5” vs 5 trap). Forgetting <code>library()</code> after <code>install.packages()</code> . Confusing parentheses () (functions) with brackets [] (positions). Swapping <code>lng/lat</code> in <code>leaflet</code> , landing the map in the ocean.

Field	Content
One activity	“Guess the picture” (10 min): students sketch where the five <code>rainfall_mm</code> dots will fall before running <code>plot(day, rainfall_mm)</code> , then explain why the dots do not form a line.
One assessment idea	Give students the volcano Day 1 / Day 2 maps and a short array-difference snippet. Ask them to predict where the maps differ, then interpret what a real one-day hot patch could mean (venting, weather, time of day) and what evidence would confirm it. Grade the reasoning, not the syntax. (This uses Part C, which is optional; teach the array first if you assess it.)

Faculty Preparation Notes

These notes are written for an instructor who knows Earth or sustainability science well but is not a confident R programmer. You do not need to master R to teach Chapter 1. You need to run the chapter’s code once before class and know what to say when something does not appear. Chapter 1 is the Tool rung: the goal is for students to feel comfortable opening RStudio and reading code, not to memorize syntax.

If you read only two things here, read the Troubleshooting table (the errors that actually happen) and the Minimum viable lesson plan (a ready-to-run 50-minute class). Both are below.

Approximate preparation time

Task	First time you teach it	Once you have taught it before
Install R + RStudio on your machine	20 min	already done
Run the whole chapter once, top to bottom	25 min	10 min
Pre-render and screenshot the 3 key figures	10 min	5 min
Skim the Cheat Sheet + these notes	10 min	5 min
Total	~65 min	~25 min

The chapter uses only base R plus two packages (`maps`, `leaflet`). There are no live internet downloads in Chapter 1, so nothing can fail because a server is down. This is the safest chapter in the book to teach. The first download-dependent chapter is Chapter 2.

Concepts to review beforehand (with book pointers)

You are reviewing the idea each piece answers, not the syntax. R is the tool.

Concept	Plain-words version to have ready	Book pointer
Object + assignment <code><-</code>	“A labeled box that stores a value so we reuse it.”	“What Is an Object?”, <code>x <- 10</code>
Vector <code>c()</code>	“A short list of numbers (or text) that belong together.”	“Vectors”, <code>rainfall_mm</code>
Indexing <code>[]</code>	“Pick a value by its position; position 1 is first.”	“Indexing in vectors”
Function + argument <code>()</code>	“A ready-made tool; the input goes in the parentheses.”	“Functions”, <code>mean()</code> , <code>sqrt()</code>
<code>summary()</code> , <code>mean()</code> , <code>length()</code>	“Quick descriptions of a vector: <code>length()</code> counts the values, <code>mean()</code> averages them, <code>summary()</code> reports smallest, middle (median), average, and largest. Tell students they can ignore the quartile (1st Qu./3rd Qu.) columns for now.”	“Simple summary”, “Vectors” (<code>length(temperature)</code>)
Data frame, <code>\$</code> , <code>head</code> , <code>str</code> , <code>names</code>	“A table like a spreadsheet; <code>\$</code> pulls one column. <code>str()</code> also reports each value’s type (<code>chr</code> = text, <code>num</code> = number, <code>logi</code> = TRUE/FALSE), which is how you diagnose a type mismatch (text where you expected numbers).”	“Data Frames”
Package vs library	“Install once; load every session. The double colon <code>::</code> calls one function straight from a package (<code>maps::map()</code> reads as ‘use <code>map()</code> from <code>maps</code> ’), which matters when two packages share a function name; typing <code>maps::</code> then Tab lists that package’s functions.”	Part B intro, <code>install.packages</code> vs <code>library</code>
<code>maps</code> outline <code>maps</code>	“Draw country/state boundaries to preview spatial data.”	<code>@fig-us-map</code> , <code>@fig-greece-map</code>
<code>leaflet</code> (interactive, optional)	“R can make a zoomable web map in a few lines.”	<code>@fig-interactive-map</code>

Concept	Plain-words version to have ready	Book pointer
List <code>[[]]</code> , array, <code>image/contour</code> , for loop	“Hold mixed or multi-layer data; repeat a task automatically.”	Part C
Reading code inside-out	“Start deepest inside the parentheses, work outward.”	“Reading and Interpreting Code”

Likely student questions (and a one-line answer)

Student asks	Your one-line answer
“What is the difference between <code><-</code> and <code>=</code> ?”	“Both assign; <code><-</code> is the R house style, use it.”
“Why <code>()</code> sometimes and <code>[]</code> other times?”	“ <code>()</code> runs a function; <code>[]</code> picks a position.”
“What is the <code>[1]</code> in the output?”	“R’s label for the position of the first value; you never type it.”
“Do I have to memorize all these functions?”	“No. The chapter says so explicitly. You learn to read, not recall.”
“Why does the map need lowercase <code>'new jersey'</code> ?”	“The dataset stores names in lowercase English; it must match exactly.”
“Is the interactive map on the exam?”	“No, <code>leaflet</code> is a preview, optional and not assessed.”
“My plot did not show up.”	“Run the <code>plot()</code> line itself; the graph appears in the Plots pane, bottom right.”
“What are the 1st Qu. and 3rd Qu. in <code>summary()</code> ?”	“Quartiles (the values one quarter and three quarters of the way through the sorted data); you can ignore them for now.”
“How do I look up what a function does?”	“Type <code>?mean</code> (or <code>?functionname</code>) at the Console; the help page, with examples near the bottom, opens in the Help pane (bottom right). For a whole package use <code>help(package = 'maps')</code> .”

Troubleshooting: the errors that actually happen in Chapter 1

These are the specific failures this chapter produces, with the exact fix to say out loud. Pre-running the chapter once means you will have seen all of them already.

Symptom on screen	Cause	Fix to say out loud
could not find function "map"	<code>library(maps)</code> not run	“Load the package first: run <code>library(maps)</code> , then re-run the map line.”

Symptom on screen	Cause	Fix to say out loud
Error ... nothing to plot or empty map for a state	Region name not lowercase English	“Names must be lowercase: use 'new jersey', not 'New Jersey'.”
Console shows a lone + and seems frozen	Unfinished command (missing) or + left dangling)	“Click the console, press Esc, then finish the parenthesis.”
object 'rainfall_mm' not found	The vector line was not run before <code>plot()</code>	“Run the lines in order, top to bottom; the box must be filled first.”
Numbers came out as text in quotes ("5")	Mixed text and numbers in one <code>c()</code>	“A vector holds one type; keep all-numbers or all-text.”
<code>mean()</code> suddenly broken	Student typed <code>mean <- 10</code> , overwriting the function	“Restart R (RStudio menu: Session then Restart R, or Ctrl/Cmd+Shift+F10); avoid naming objects <code>mean</code> , <code>data</code> , <code>plot</code> .”
Map of a state appears but is just an outline	Expected; no <code>fill/col</code> given	“That is correct, it is an outline. Add <code>fill = TRUE</code> , <code>col = '...'</code> to shade it.”

What can be safely skipped if time is limited (Part A/B/C tiering)

The chapter is already tiered. Use that.

Tier	Sections	Decision
Must-teach	Part A: objects/ <code><-</code> , <code>c()</code> vectors, <code>[]</code> indexing, the rainfall scatter (<code>@fig-first-plot</code>), functions/arguments, data frames + <code>\$/head/str/names</code>	Teach in full. This is the foundation every later chapter assumes.
Recommended	Part B: packages vs libraries, <code>library(maps)</code> , the US/Greece/NY/NJ outline maps	Teach the package idea and one or two maps. Maps preview Chapter 3 (spatial) and motivate “why code for Earth data.”
Optional	Part C: lists <code>[[]]</code> , arrays + <code>image/contour</code> , for loops, the volcano Day 1 / Day 2 application (<code>@fig-volcano-temp</code>); also the <code>leaflet</code> interactive map in Part B	Skip or assign as reading if short on time. Flag the volcano map as a preview, not a test item. The book itself labels Part C optional.

The `leaflet` step is interactive and optional. Label it that way when you show it. It is a “look what R can do” moment, never a graded skill.

Minimum viable lesson plan (one 50-minute class)

If you have a single session and limited confidence, teach only this. Every step has a fallback so a projector hiccup never stalls the class. This is the compressed path (it trims the indexing and `str()` detail and includes only one map); the fuller Part A treatment runs about 75 minutes, see Estimated teaching time.

Min	Do	Say / fallback
0-5	Open RStudio; point at Script (top-left) and Console (bottom-left)	“Script = notebook you keep; Console = scratch paper.”
5-12	<code>x <- 10; then day <- 1:5</code> and <code>rainfall_mm <- c(0,12,5,0,8)</code>	“We are storing Earth measurements in labeled boxes.”
12-22	Run <code>plot(day, rainfall_mm)</code> (@fig-first-plot)	Ask “is rainfall rising or falling across the days?” before discussing. Fallback: show the pre-rendered screenshot.
22-32	Build the 3-station <code>data.frame</code> ; run <code>head(df)</code> , <code>df\$temp</code>	“Real Earth data look like this table: stations down, measurements across.”
32-42	<code>library(maps)</code> then <code>map("state")</code> and one state, e.g. New York	“We can preview where our data live. Watch the lowercase name.” Fallback: pre-rendered map image.
42-50	Recap with “Key Takeaways”; assign the chapter Exercises 1-5	“You read code and a plot today. That is the whole job: ask a question, see the answer.”

What to skip in the minimum plan: all of Part C (lists, arrays, loops, volcano maps) and the `leaflet` preview. Mention they exist as optional reading. Nothing later in the book breaks if you defer them.

Narrative, mindset and ladder position

Chapter 1 is the **Tool rung**, the bottom of the Analytical Ladder. Every later chapter (trustworthy data, space, time, relationships, Earth systems, sustainability) stands on it. Your teaching job here is narrow and important: get a nervous, code-free beginner to the point where R runs, a result appears, and the result can be questioned. Nothing more. The science deepens later; this rung removes the barrier to entry.

Came from. There is no prior rung. The honest recap is: “You have no coding background, and that is fine. The book says so on its first page.” The book’s own framing, “students do not struggle with complexity; they struggle with abstraction,” is your north star. Keep everything hands-on and concrete.

The new problem this rung answers. Earth data are everywhere (rain gauges, wells, satellites, GPS), but a beginner has no way to open them, look at them, or repeat the look tomorrow. A graphical menu can show one map once. It cannot guarantee the same result next year on another computer. This rung hands the student a tool that can.

Why this rung exists first. You cannot ask “can I trust these numbers?” (Chapter 2) until you can load and see numbers at all. R is introduced here purely as the vehicle that carries every later question.

Hand-off to the next rung. End the chapter by naming Chapter 2’s question out loud: “Now that R runs and you can see a plot, the real question becomes: can you trust the data behind the plot? In Chapter 2 you load a real USGS groundwater record, gaps and all.” That sentence keeps the climb feeling continuous.

Mindset to set, in your own voice:

- R is the tool, never the topic. If a class minute is spent on syntax for its own sake, redirect it to “what is this number telling us about the Earth?”
- *Think twice, code once* (the book’s guiding phrase). Predict before you run.
- Reading code well matters as much as writing it. The chapter teaches reading inside-out deliberately.
- Errors are normal and expected. A frozen + prompt is a missing parenthesis, not a failure. Model calm recovery (Esc) on day one.

Sustainability Lens

The Tool rung carries a quiet sustainability message that is easy to skip and worth stating to students directly:

- **Reproducibility is an ethical and sustainability skill, not a convenience.** A water-board decision, a flood warning, or an emissions report is only trustworthy if someone else can re-run it and get the same answer. A script does that; a sequence of menu clicks does not. The book makes this its opening argument (“from clicking to thinking”), and this is where students first feel it.
- **The data structures preview real Earth systems.** A vector is a rain record. A data frame is a network of monitoring stations. An array is a stack of maps changing over time, exactly the shape of satellite or temperature data students will analyze for climate and groundwater questions later. Frame each structure as “this is how a real Earth dataset is shaped,” not as “this is an R container.”
- **Scope check for instructors:** keep the stake small and tangible here. Five days of rain that could flood a street, one volcano with a localized hot vent. Save global emissions and aquifer depletion for the chapters built for them. An honest small anchor beats a forced global one.

A. Orient

Learning objectives

State these as thinking outcomes. By the end of Chapter 1, a student can:

#	Thinking outcome (not a syntax outcome)
1	Explain, in plain words, why coded analysis is repeatable and a menu workflow is not, and why that matters for an Earth or sustainability result.
2	Predict the rough shape of a simple plot from the numbers alone, then reconcile the prediction with what R drew.
3	Read a nested line of code from the inside out and say what each layer produces before running it.
4	Match a data structure to the shape of a real Earth dataset (a record to a vector, a station table to a data frame, a time-stack of maps to an array).
5	Look at a result (a plot, a difference map) and ask “does this make sense for the real system?” rather than “did the code run?”

Deliberately **not** an objective: memorizing function names or argument lists. Tell students this explicitly; it lowers anxiety and keeps attention on thinking.

Prerequisite knowledge

Students need	Students do NOT need
R and RStudio installed (book gives the steps and order). To find the Script pane and the Console. The keyboard run shortcut (Ctrl+Enter / Command+Enter). Comfort that millimetres measure rainfall and that a coordinate is a place.	Any prior programming, in any language. GIS, statistics, or Earth-science coursework. To know what a package, vector, or array is (taught here). Calculus, linear algebra, or the meaning of quartiles (the book flags these as “not needed for now”).

If a student arrives without R installed, that is a setup task, not a learning task. Resolve it before the analytical content begins so the first hands-on minute produces a result, not a download.

Estimated teaching time

Coverage	In-class time	Notes
Part A only (objects, vectors, indexing, functions, data frames)	~75 min	Sufficient for a beginner foundation; the book says Part A alone builds a strong base.
Parts A + B (adds packages, maps, leaflet preview)	~110 min	The natural one-week target for most courses.
Parts A + B + C (adds lists, arrays, loops, volcano application)	~150 min, or split over two sessions	Part C is optional; the volcano maps are a strong payoff if time allows.
leaflet interactive preview	5 min, demo only	Label it optional/interactive; do not assess it.

Budget more time than the code suggests. The minutes go to prediction and interpretation, not typing.

Common student misconceptions

Misconception	Why a beginner holds it	Quick correction
"Learning R is the goal of the course."	It is the first visible skill.	R is one tool among many (menu, Python). The goal is reading Earth data. Say this on day one and repeat it.
Dots on a plot should connect into a line.	School graphs often connect points.	Each dot is one independent measurement (one day's rain). The up-and-down shape is the data speaking.
"5" and 5 are the same.	They look identical.	One is a quantity R can compute with; one is a label (like a ZIP code). The book's "burger vs picture of a burger" analogy lands well.

Misconception	Why a beginner holds it	Quick correction
A plot that “looks rotated” is a mistake.	The image() grid differs from the printed table.	image() draws like a map (bottom-up); the Console prints like a spreadsheet (top-down). Same values, different orientation. (R also fills an array column by column, so 1, 2, 3 run down the first column first; this is why the grid layout can surprise beginners.)
Code must be memorized to be useful.	It feels like a foreign language.	The skill is reading and predicting, not recall. Reassure repeatedly.

Common coding mistakes

These are the friction points that derail a first session. Pre-empt each; do not let them become the lesson.

Mistake	Symptom students see	One-line fix to teach
Working only in the Console.	Work vanishes when R closes.	Write in a Script, run with Ctrl+Enter.
Incomplete command (5+).	A stuck + prompt waiting.	Click the Console, press Esc, finish the line.
Missing closing).	Error, or the + prompt again.	R reads until it finds); check parentheses match.
install.packages() vs library() confusion.	“Could not find function”.	Install once (with quotes); load every session (no quotes).
Reserved names (mean , data , plot).	Strange downstream errors.	Use descriptive names (mean_value , rain_mm).

Keep the framing positive: every item above is a normal beginner moment with a one-step recovery, not a sign of failure.

B. Teach

Lecture outline

A reading-first, syntax-light sequence. The bolded line is the question you lead with; the code serves it.

1. **“Why code at all, when a menu is easier?”** Open on reproducibility (Preface argument). A coded analysis re-runs identically; a menu workflow drifts. (5 min, no code.)
2. **“How do I store a number so I do not retype it?”** Objects and `<-`. Anchor: store one temperature you will reuse. (Objects.)
3. **“How do I see a pattern in a few numbers?”** Vectors with `c()`, then `plot(day, rainfall_mm)` (@fig-first-plot). Predict the shape first. (Vectors + first plot.)
4. **“How do I pull out the value I want?”** Indexing with `[ ]`; `contrast()` (functions) vs `[ ]` (positions). (Indexing.)
5. **“What is a tool that does work for me?”** Functions: input to output; read `mean(c(...))` inside-out. If time allows, write one tiny function (`celsius_to_fahrenheit`) to make “input to output” concrete; otherwise just read one. (Functions.)
6. **“How is a real station dataset shaped?”** Data frames (`data.frame`, `$`, `head`, `str`, `names`) as a spreadsheet of stations. (Data frames.)
7. **“How do I add tools R does not ship with?”** Packages: `install.packages` vs `library`, `::`; draw US / Greece / NY / NJ outlines with `maps`. (Packages.)
8. *(Optional, interactive)* **“Can R make a zoomable web map?”** leaflet preview. Demo only; label optional. (5 min.)
9. *(Optional, Part C)* **“How is Earth data that changes over space and time shaped?”** Lists, arrays, loops, ending on the volcano Day 1 / Day 2 temperature maps (@fig-volcano-temp). (Arrays as map-stacks.)
10. **Close: the hand-off.** “R runs. Now, can you trust the data? That is Chapter 2.”

Analogies

The book ships several strong analogies; reuse them rather than inventing new ones, and add the Earth framing.

Concept	Analogy (use verbatim or close)
Object / <code><-</code>	A labeled box: <code>x</code> is the label, 10 is what is inside.
Script vs Console	Notebook (saved) vs scratch paper or calculator (gone when you close).
"5" vs 5	A real burger you can eat vs a picture of a burger.
Vector	A short record that belongs together: five days of one rain gauge.
Data frame	A spreadsheet of monitoring stations, one row per station.
Array	A stack of maps , one layer per day, the shape of satellite or temperature data over time.
Package	R base is the core toolkit; a package is an extra toolbox; functions are the tools inside.
Reading inside-out	Open the innermost box first (<code>c(...)</code>), then the box around it (<code>mean(...)</code>).
Multi-line function call	R reads a function until it meets the closing <code>)</code> , so a <code>c(...)</code> split across several lines is one command; place the cursor on any line inside it and run.

Real-world hooks

Every concept gets an Earth anchor. Lead with the stake, then the code.

Concept	Earth/sustainability hook
Vector + scatter (<code>rainfall_mm</code>)	Five days of rain that decide whether a storm drain copes or a street floods. Timing of the rain may matter more than the total.
Mixing text and numbers	A ZIP code, a USGS station ID, an earthquake event ID: looks numeric, but is a label. You never average station IDs.
Data frame	A network of weather or groundwater stations, one row per station, exactly Chapter 2's USGS shape.
maps (US/Greece/NY/NJ)	"Where does this happen?" The first taste of the Space rung: putting a phenomenon on a map.
Array (volcano Day 1 / Day 2)	A localized hot vent (fumarole) that warms one patch overnight while the rest of the field stays the same: how a real Earth surface changes in space and time.
Loop over layers	100 satellite images over 100 days: you cannot hand-plot each one.

Active-learning exercises

Each follows predict, then run the smallest real code, then interpret. The prediction must be committed (on paper or in chat) before any code runs. Interpretation must be in Earth terms, not code terms.

10-minute: "Guess the picture" (the rainfall scatter, `@fig-first-plot`).

- *Predict (3 min, no R)*: "Five days of rain in mm: 0, 12, 5, 0, 8. Sketch where the five dots sit. Which day is highest? Which sit on the zero line? Will they form a straight line?"
- *Run (2 min)*:

```
day <- 1:5
rainfall_mm <- c(0, 12, 5, 0, 8)
plot(day, rainfall_mm)
```

- *Interpret (4 min)*: "The dots do not form a line. Why not? What would have to be true about the rain for them to?" Target insight: each dot is one independent day; the up-and-down is the weather, not noise.
- *What if (1 min, fast finishers)*: "Change day 4 from 0 to 20. Re-predict which dot moves before re-running."

20-minute: "Read the station table" (data frame).

- *Predict (5 min, no R)*: Show the three-station data frame in slides. "Which station is warmest? How many rows will `head(df)` show? What will `df$temp` give back, one number or three?" Commit answers.
- *Run (5 min)*:

```
df <- data.frame(
  station = c("A", "B", "C"), # station names
  temp = c(15.2, 16.4, 14.8), # temperature values
  rain = c(2.1, 0.0, 5.4)      # rainfall values
)
head(df)
df$temp
```

- *Interpret (7 min)*: “Which station would you visit first if you were studying a heat anomaly, and why? What does `$temp` returning three values tell you about how the table is organized?”
- *What if (3 min)*: “Add a fourth station with a missing temperature. Predict what changes before you run it.” (Previews Chapter 2’s missing-data theme.)

50-minute: “How does a volcano surface change in a day?” (array + the Day 1 / Day 2 maps, @fig-volcano-temp). Three linked predict-run-interpret cycles, ending in a small conclusion. Pre-load `volcano` so no setup eats the clock. Before class, run this whole block once and screenshot Day 1, Day 2, and the difference map; if anything misbehaves live, project the screenshots and keep teaching. Every step has a fallback (see the Troubleshooting table in Faculty Preparation Notes).

- *Cycle 1, structure (15 min)*: Predict: “We will stack two days of a temperature field as `array(NA, c(nrow(volcano), ncol(volcano), 2))`. How many layers? What does each layer represent?” Run the array setup. Interpret: “Why store two days as a stack instead of two separate tables?”
- *Cycle 2, the change (20 min)*: Predict, on paper, where on the map the two days will differ, before seeing it. Then run the `image()` + `contour()` Day 1 / Day 2 side-by-side block. Interpret: “Where did the surface warm? The rest looks identical. In the real world, what could heat one small patch overnight (a vent, weather, time of day)? What evidence would confirm it?”
- *Cycle 3, the difference (15 min)*: Switch to the simpler demo array from Part C, `arr <- array(1:24, dim = c(3, 4, 2))`, whose two layers differ by exactly 12 everywhere. Predict the look of `image(arr[, , 2] - arr[, , 1])`, run it, and interpret why a uniform color means “same change everywhere,” and why real data almost never looks that clean. (Note for you: this `arr` is a different, simpler array than the volcano `temp`; the volcano difference would be patchy, not uniform, which is exactly the contrast worth drawing out.) Group share-out: one sentence each on “what a difference map tells an Earth scientist.”

For all three, grade the written prediction and interpretation, not whether the code ran.

Think-pair-share

Short, reasoning-first prompts. Think alone (1 min), pair (2 min), share (2 min).

1. **Reproducibility.** “Your lab mate did a flood analysis by clicking menus and cannot reproduce it six months later. You did it in a script. Whose result would a city water board trust, and why?”
2. **Label vs quantity.** “Is a groundwater station ID a number or a label? Pair up and list three Earth IDs you should never do arithmetic on.”

3. **Structure choice.** “You have temperature at many locations changing daily. Vector, data frame, or array? Defend your choice to your partner.” (Target: array, a stack of maps over time.)
4. **Reading inside-out.** “Without running it, talk through `mean(c(10, 12, 14, 16))` from the inside out. What is built first, what happens second, what prints?”

Discussion questions

Whole-class, open-ended, interpretation-weighted. None has a one-word syntax answer.

1. The Preface argues coding beats menus for serious work, yet admits menus are “perfect” for quick field decisions. When would you still reach for a menu, and when does that choice become a problem for a result others must trust?
2. The book repeats “R is not the goal.” If R is only a tool, what is the goal of this course in one sentence, and how would you tell whether you achieved it without writing any code?
3. The rainfall example uses five made-up days. What is gained, and what is lost, by starting with small invented numbers instead of a real download? When should a teaching example “grow up” into real data?
4. The volcano Day 2 map shows one warm patch. List two explanations from the Earth (a vent, weather) and one from the data itself (a sensor error). How would you decide between them? (This rehearses the trustworthy-data judgment that defines Chapter 2.)
5. Looking up the ladder: now that you can load and plot data, what is the very next question you would want to ask of an Earth dataset, and which later chapter do you think will answer it?

C. Interpret

This chapter is the Tool rung: students are learning what R is, not yet analyzing real Earth data. That makes its two figures unusually risky to teach. Both are toy datasets (five invented rainfall values; a built-in `volcano` grid relabelled as temperature), so the single most important interpretive lesson of Chapter 1 is the one students least expect: **a figure can be perfectly real-looking and still support almost no scientific conclusion.** Use these figures to install the habit (“Does this result make scientific sense?”) on data simple enough that the honest answer is “barely.” Then Chapter 2 onward earns the harder conclusions.

Teach the reasoning, not `plot()`, `image()`, or `contour()`. The code is covered in the chapter body; do not re-explain it here. Point students back to it (`@fig-first-plot`, `@fig-volcano-temp`) and spend class time on what the picture does and does not mean.

Instructor note on units and subscripts: when you reproduce these figures on slides or handouts, write prose subscripts in Pandoc markup (`C0~2~`, `log~10~`), never as Unicode glyphs, and keep degree units as `°C` / `°F`. Rainfall is in `mm`; the volcano values are unit-free grid numbers we are pretending are temperatures.

C.1 The daily-rainfall scatter (@fig-first-plot)

Five points: `day` (1 to 5) on the horizontal axis, `rainfall_mm` (0, 12, 5, 0, 8) on the vertical axis. Two days sit on the baseline at zero.

Layer	For this figure
Notice first	The five dots rise and fall across the days. Nothing is flat; two dots sit right on the bottom.
Notice second	Only five points exist (far too few to call anything a pattern). The two zeros are real measured dry days , not missing data. The vertical axis spans only 0 to 12 mm , so small differences look dramatic. The horizontal axis is day number, not a calendar, so we do not even know the season or the year.

Justified vs NOT justified

Justified (the figure supports this)	NOT justified (the trap)
Rainfall differed from day to day across these five days.	“Rainfall is increasing or decreasing.” (5 points is not a trend.)
Two of the five days had no rain (a true zero).	“This is the rainy or dry season.” (No calendar, no location.)
A picture reveals the up-and-down shape faster than reading the four numbers.	“Day 6 will be higher.” (Extrapolation beyond the data.) “Rain here averages about 5 mm/day .” (The mean of 5 invented numbers describes nothing real.)

Common misinterpretation (pre-empt it out loud): students read the zero-rain days as missing or broken data. Correction: here 0 is a genuine measurement (it did not rain), and that contrast is worth stating now, because in Chapter 2 (@fig-gw-raw) a blank gap will mean the opposite (no measurement, value unknown / NA). Saying “zero is a value; a gap is the absence of a value” on this easy figure pays off directly at the next rung.

Interpretation discussion questions (none answerable by re-describing the dots):

1. How many days of rainfall would you need before you would feel comfortable using the word “trend,” and why is five not enough?
2. The chart shows day numbers, not dates. What would you need to know before claiming anything about seasons or climate?
3. A friend says “it rained more later in the week.” Point to exactly which dots support that, then explain why the claim still might not generalize beyond these five days.
4. If one zero had instead been a blank (no dot at all), how would your reading change? (Sets up missing data in Chapter 2.)

C.2 The volcano Day 1 / Day 2 temperature maps (@fig-volcano-temp)

Two terrain-coloured maps side by side with contour lines (isotherms), built from the built-in `volcano` grid, relabelled as temperature, with a small square in the north of Day 2 raised by +50.

Say this first, every time you show it: this is simulated and illustrative, not measured field data. `volcano` is a topography grid we are treating as if it were temperature, and the hot patch was added by us on purpose. Most over-claims here come from students forgetting the data are invented.

Layer	For this figure
Notice first	Two almost-identical coloured maps with squiggly contour lines; Day 2 has a small bright or hot patch in the north.
Notice second	The two maps are the same field except for one block we edited; the change is localized in space and across time (one spot, one day later). Contour lines = lines of equal value (isotherms), so tightly packed lines mean a steep change, widely spaced lines a gradual one. The colour scale runs light-to-dark with value; check the legend direction before claiming “red = hot.” The grid has no real units, no real location, and only two time steps .

Justified vs NOT justified

Justified (the figure supports this)	NOT justified (the trap)
Between the two layers, one localized area in the north changed while the rest stayed essentially the same.	“The volcano is warming or about to erupt.” (Invented data; nothing physical is measured.)
Earth data can change across space AND time at once , and a stacked array can hold that.	“Temperatures rose 50 degrees.” (Unit-free grid; the +50 is an illustrative number, not °C.)
Contour lines let you locate where values are similar and where they change steeply, without reading every cell.	“The whole map heated up.” (Only the edited square changed.)
Side-by-side maps make a small local change easy to spot.	“The pattern of this volcano applies to volcanoes generally.” (One simulated grid.)

Common misinterpretation (pre-empt it out loud): students treat the second map as real evidence of volcanic heating over time and start reasoning about eruptions. Correction in one sentence: we typed the hot spot in ourselves; the figure demonstrates how to represent and visualize spatial change over time, it does not detect any. A close second misread is colour direction: confirm

from the legend whether dark or light is the high value before anyone says “the red shows the heat.”

Interpretation discussion questions (require reasoning, not description):

1. If this were real measured data, what would you need (units, instrument, more than two days, more than one volcano) before claiming the volcano is heating up?
2. The contour lines are denser in some places than others. What does spacing between contour lines tell you about how fast the value changes across the surface?
3. Day 1 and Day 2 differ in exactly one square. In a real monitoring dataset, what natural causes other than genuine heating could make one small area look different between two days?
4. Why is “two time steps” almost never enough to claim a change is a trend rather than a one-off? (Connects to the rainfall figure and forward to the time-series chapters.)

C.3 Instructor checklist (use before teaching either figure)

- ☐ Said out loud that **both datasets are invented or illustrative**, so the lesson is how to read and represent data, not what these specific data prove.
- ☐ Distinguished **zero-as-measurement** (rainfall) from **gap-as-missing** (NA, coming in Chapter 2) at least once.
- ☐ Named the three limits that kill most over-claims here: **tiny sample, no real units or location, no real time span**.
- ☐ Checked the **colour-scale direction from the legend** before stating what high vs low means on the volcano maps.
- ☐ Explained contour lines as **lines of equal value** (isotherms; say “lines of equal temperature” to students), with spacing = rate of change.
- ☐ Asked at least one question that **cannot** be answered by re-describing the picture.
- ☐ Kept the discussion on **Earth-data interpretation**, not on `plot()` / `image()` / `contour()` syntax (that lives in the chapter body).
- ☐ (Optional / interactive) If demoing the `leaflet` preview map live, labelled it clearly as **interactive and optional**: it shows R can make web maps; it is not a result to interpret.

D. Practice

This section is teaching material for the instructor. Chapter 1 is the Tool rung of the book’s Analytical Ladder: students are learning to store Earth data in R structures (vectors, data frames, lists, arrays) and to read code inside-out, before they download real USGS data in Chapter 2. So every exercise below is framed as predict, then run, then interpret, not “type this and submit a screenshot.” The deliverable you grade is the student’s reasoning about Earth data, not their R syntax. Point students back to the relevant chapter section rather than re-teaching it here.

How the three blocks fit together:

Block	What the student does	What you are growing
Coding exercises	Predict an output, run the chapter's own code, interpret the result	Reading code as a description of an Earth dataset
Project ideas	Reuse Chapter 1 structures on data they choose	Seeing R structures as containers for real measurements
AI activities	Use AI to check, challenge, or question their thinking	Judging an answer instead of accepting it

Coding exercises

Use only base R plus the `maps` package (and the optional `leaflet` preview). No new functions are needed. Every exercise asks the student to **commit to a prediction in writing before running**, then **interpret after**. Grade the prediction-and-interpretation, not whether the code ran.

Note

Two arrays appear in Part C, keep them straight. `arr` is the small 1:24 demo array (its Day 2 minus Day 1 difference is a flat 12 everywhere, `@fig-array-diff`). `temp` is the `volcano` array (its difference is a single northern patch, `@fig-volcano-temp`). The exercises below name which is which; copying `arr` code expecting the volcano result is the most common slip.

How to assign each one (instructor pattern):

- ☐ Student writes their prediction first (one or two sentences).
- ☐ Student runs the chapter's code, unchanged or lightly edited.
- ☐ Student compares output to their prediction.
- ☐ Student answers the Earth-data interpretation question.

#	Anchor in Ch. 1	Predict (before running)	Run	Interpret (the graded part)
1	The rainfall scatter, <code>@fig-first-plot</code> (<code>rainfall_mm <- c(0, 12, 5, 0, 8)</code>)	"Which two days will sit on the bottom axis, and why?"	<code>plot(day, rainfall_mm)</code>	Two days read 0 mm. Are those true dry days, or could a 0 mean a sensor that failed to report? What evidence would tell the two apart?

#	Anchor in Ch. 1	Predict (before running)	Run	Interpret (the graded part)
2	The <code>temperature</code> vector and indexing	“What will <code>temperature[2:4]</code> return, and how many values?”	<code>temperature[2:4]</code>	If these were four daily readings, what does selecting positions 2 to 4 mean in time? When would an analyst want a slice like this?
3	The <code>df</code> data frame (<code>station</code> , <code>temp</code> , <code>rain</code>)	“What will <code>df[, 2]</code> show, and what does each number represent?”	<code>df\$temp</code> then <code>df[, 2]</code>	The column is temperature per station. If station B read 0.0 rain, is that a dry station or a missing value entered as zero? Why does that distinction matter before any analysis?
4	<code>summary(values)</code> on a vector	“Will the mean be above or below the middle value? Guess before running.”	<code>mean(values)</code> , <code>summary(values)</code>	If <code>values</code> were daily temperatures, what would a mean far from the median hint at (one extreme hot day, a station error)?
5	The <code>maps</code> outline maps (<code>@fig-ny-map</code> , <code>@fig-nj-map</code>)	“Will <code>map('state', regions = 'New Jersey')</code> work as written?”	Run it, then fix it to <code>"new jersey"</code>	The fix is lowercase matching. Connect it to real data: station IDs and place names must match the dataset’s spelling exactly or the join fails. Where else will this bite them later?

#	Anchor in Ch. 1	Predict (before running)	Run	Interpret (the graded part)
6	The <code>volcano</code> array <code>temp</code> , Day 1 / Day 2 maps, <code>@fig-volcano-temp</code>	“After raising rows and columns 48:50 of <code>player</code> 2 by 50, how much of the Day 2 map changes?”	Re-run the chapter’s volcano array (<code>temp</code>) and image code	Only a small northern patch changes. What real Earth process could heat one small area between two days (a vent, a fire, an instrument drift) and leave the rest unchanged?
7	The array difference image, <code>@fig-array-diff</code> (the 1:24 demo array <code>arr</code> , not the volcano)	“What color pattern will <code>image(arr[, , 2] - arr[, , 1])</code> show?”	Run it	The image is one flat color because every cell differs by exactly 12. When would a real difference map be flat, and would a flat difference make you trust or distrust the data?
8	The <code>for</code> loop over the 1:24 demo array (<code>arr</code>) layers, <code>@fig-loop-layers</code>	“How many images will the loop draw, and why does <code>i</code> matter?”	Run the 2-layer loop	If <code>arr</code> held 100 daily layers instead of 2, the same loop draws 100 maps. What is the analyst really automating here: typing, or a repeated scientific comparison?

Stretch (transfer item, not in the chapter; still base R + maps): give the student a vector `co2_ppm <- c(410, 412, 414, 417, 420)` and ask them to predict the trend, plot it with `plot()`, and say in one sentence what a rising CO₂ series means for the planet, and what they would need to claim it is a real long-term trend rather than five noisy points.

Project ideas

Each project reuses a Chapter 1 structure on data the student gathers or invents from a real place, so the structure stops feeling abstract. None requires a package beyond `maps`. The output is a short written interpretation, not a code dump.

Project	Ch. 1 structure reused	The Earth-data question (the point)
My local week of weather	<code>vectors + plot()</code> (<code>@fig-first-plot</code>)	Record 7 days of rainfall or temperature for your town. Plot it. Which day stands out, and is it weather or a reading error?
Three stations table	<code>data.frame</code> , <code>\$</code> , <code>head()</code> , <code>str()</code>	Build a 3-station table (real or plausible) for one region. Use <code>str()</code> to confirm each column's type, then argue which column is a label (like a station ID) and which is a quantity.
Map my study area	<code>maps::map()</code> with <code>regions=</code>	Draw the state or country you would study. Write two sentences on what a question at this place might be (coastal flooding, groundwater, emissions).
A two-day change map	3D array + <code>image()</code> + <code>contour()</code> (<code>@fig-volcano-temp</code>)	Invent a small grid for “before” and “after” an event (a heatwave, a melt). Add change to one region only. Interpret where and why it changed.
Station notebook as a list	<code>list</code> , <code>\$</code> , <code>[[]]</code>	Store one station's metadata (name, coordinates vector, a small measurements data frame) in a single list. Explain why a list, not a data frame, fits mixed information.
Interactive (optional)	<code>leaflet</code> preview (<code>@fig-interactive-map</code>)	Interactive, optional. Center a <code>leaflet</code> map on your study site (<code>lng</code> then <code>lat</code>). Confirm it does not land in the ocean, and note one thing a zoomable map shows that a static outline cannot.

Grading note: reward the question and the interpretation. A rough plot with a sharp, honest reading (“I think day 3 is a sensor gap, not a dry day”) beats a clean plot with no thought.

AI activities

Ground rule for students (hand this out): AI is a tutor and a sparring partner, never a ghostwriter. In Chapter 1, AI may explain a result, question your reading, or generate practice. It may **not** write the code you submit or interpret the figure for you. Every activity below pairs a **GOOD** prompt with the **BAD** prompt students are tempted to type, plus the one-line reason and a verify step. The skill behind these: the student commits first, AI responds second, the student checks AI against the chapter last.

Activity 1: Reflection on the rainfall scatter (@fig-first-plot). *Thinking target: explain in your own words why a result makes scientific sense.*

	Prompt
BAD	“Explain everything about this rainfall plot.”
GOOD	“I think the two points sitting on the bottom axis are days with 0 mm of rain. Tell me whether that reading is right, then ask me what a point for a 20 mm day would look like. Do not give me code.”

Reason: the BAD prompt produces a wall of text the student skims; the GOOD prompt makes the student commit to a reading first, then checks it. **Verify:** look back at `rainfall_mm <- c(0, 12, 5, 0, 8)` and confirm which days are actually zero, instead of trusting the AI’s count. **Reflect (own words):** “Did the AI agree with my reading? Could a 0 mean ‘no rain’ or ‘no data’, and how would I tell?”

Activity 2: Debugging-as-understanding with maps (@fig-nj-map). *Thinking target: read an error and reason about its cause before fixing.*

The student runs `map("state", regions = "New Jersey")` (capitalized) and gets an error.

	Prompt
BAD	“Fix my map code.”
GOOD	“I ran <code>map('state', regions = 'New Jersey')</code> and got an error. Do not fix it yet. In one sentence, tell me what the error says about how my region name compares to the dataset’s names.”

Reason: the BAD prompt hands back working code with nothing learned; the GOOD prompt makes the student read the error, the skill that transfers to every dataset mismatch later. **Verify:** run `map("state", plot = FALSE)$names` and find the exact spelling the dataset expects, then fix it themselves. **Reflect:** “Where else in Earth data (station IDs, country names) will an exact-match rule like this matter?”

Activity 3: Interpretation + evaluate-the-AI-answer on the volcano maps (@fig-volcano-temp). *Thinking target: move from describing the maps to a justified conclusion, and judge the AI.*

	Prompt
BAD	“Write R code for two volcano temperature maps with a hot spot.”
GOOD	“Here are two maps of the same volcano on Day 1 and Day 2. I think the only difference is a small hot patch in the north. Ask me three questions that test whether I understand why that patch appears, then wait for my answers before explaining.”

Reason: the BAD prompt hands the student finished code to copy; the GOOD prompt keeps the student doing the interpreting while AI probes their understanding. **Verify:** re-read the chapter’s own line that raises rows and columns 48:50 of layer 2 by 50, and confirm the change is only there. If the AI claims the whole map warmed, the student has caught a real error. **Reflect:** “In plain language, why does only the northern patch change, and how would the maps differ if we had heated the south corner instead?”

Activity 4: Evaluate-the-AI-answer on the array difference (@fig-array-diff). *Thinking target: judge and verify an AI claim against the data.*

	Prompt
BAD	“Is the difference map correct? Yes or no.”
GOOD	“An AI told me the difference map <code>arr[, , 2] - arr[, , 1]</code> is one flat color because ‘nothing changed.’ Here is the array code. List two things I should check before I trust that explanation.”

Reason: the BAD prompt invites blind acceptance; the GOOD prompt forces the student to verify a plausible-sounding but wrong claim (the value did change, by exactly 12 everywhere). **Verify:** print `arr[, , 1]` and `arr[, , 2]` and confirm every cell differs by 12, so “nothing changed” is false even though the image is uniform. **Reflect:** “When would a flat difference map in real Earth data make me suspicious of the sensor rather than the planet?”

Instructor gate for any AI activity you add: before assigning it, re-read it and ask, could a student finish this by pasting the prompt and copying the reply with no thinking? If yes, it has failed. Every activity must name a real Chapter 1 artifact, make the student predict or read first, and end with the student checking AI against the chapter, never the chapter against the AI.

E. Research Extension

Chapter 1 is the Tool rung: students leave it able to read code, store data in vectors and data frames, draw a base-R plot, and stack two maps in an array. That feels like a setup chapter, not a research chapter. This section exists to prove otherwise. The signature student output below is

enough to start a real investigation, and the book’s own Acknowledgments record beginners with no coding experience doing exactly that at scientific conferences and competitions.

For Chapter 1 itself, the one piece most instructors will use right away is the Conference Readiness checklist just below, as a “what good looks like” stretch rubric. The research questions, poster skeleton, and talking points that follow are there for the student who wants to run ahead, and become central in later chapters.

Conference Readiness Indicator

The signature Chapter 1 student output: a two-day change map. This is the `@fig-volcano-temp` pattern, two side-by-side fields (Day 1, Day 2) plus their difference, built from the base-R `volcano` grid stored in a three-dimensional array. If a student can produce this and say in one sentence what the warm patch means, they have everything a first poster needs: a before, an after, and a difference that answers a yes/no question. Treat this figure, not the syntax around it, as the conference-readiness milestone for Chapter 1.

A student is conference-ready on Chapter 1 when they can check every box:

Readiness check	What it proves
Produces the Day 1 / Day 2 / difference figure unaided	Owens the full array-to-image workflow, not one line of it
States the Earth question in one plain sentence	Has a research question, not a coding exercise
Names where the warming is on the map (north, a corner)	Can read a figure spatially, the core skill of a poster
Says one honest limitation out loud (“my data is simulated”)	Thinks like a scientist, not a button-pusher
Names the next dataset that would make it real (satellite thermal)	Sees the chapter output as a starting line

If a student misses the last three boxes, they have a plot; if they hit all five, they have a poster abstract.

From a Chapter 1 output to a research question. Each signature output opens a difficulty ladder, so there is an entry point for every student. Anchor every question to a real Earth process; the figure is only the doorway.

Chapter 1 output	Descriptive question	Comparative question	Explanatory / predictive question
<code>@fig-volcano-temp</code> two-day map	Where between two days does the surface warm most?	Do north-facing cells warm differently from south-facing cells?	Does the warm patch line up with a known vent or fumarole?
<code>@fig-first-plot</code> rainfall scatter	Which of the five days was wettest, and by how much?	Does a wet day tend to follow another wet day?	If we had 50 days, could we say anything about the trend?

Chapter 1 output	Descriptive question	Comparative question	Explanatory / predictive question
The maps outline maps (US / Greece / NY / NJ)	Which boundary are we even looking at, and at what scale?	Does a state outline alone tell us anything, or do we need data on top of it?	What variable could we shade this outline with to make it a finding?

Project formats, matched to a venue. State plainly which output fits which audience so the student aims at a real room of people and a real deadline.

Format	Build it from	The deliverable
Science-fair board	<code>@fig-first-plot</code> rainfall scatter, or one difference map	One clear yes/no question, one chart, a board
HUSSRP-style poster	The <code>@fig-volcano-temp</code> two-day change map	A question, methods, one main figure, an interpretation
Undergraduate project	The array workflow applied to a real before/after raster	A short reproducible script and one labeled figure
Capstone / independent study	Two-day map extended to many days of real thermal data	A persistence test: signal versus noise across dates
Conference talk	Any of the above, once the result is solid	One claim, five talking points, a few slides

Publication-quality figure suggestions. A textbook figure is rarely poster-ready. Name the one upgrade that makes it presentable; point to the book's existing palette and caption rules rather than re-teaching plotting.

- Replace the simulated **volcano** field with a real before/after thermal pair, then draw the difference with a grayscale-safe `hcl.colors(n, "YlOrRd", rev = TRUE)` ramp so the poster survives black-and-white printing.
- Add a labeled color bar in real units (temperature in °F or °C), and mark the single warmest cell.
- For the rainfall scatter, label axes with units (day number, rainfall in mm) and add a second site or a second week so there is something to compare.
- For a **maps** outline, shade it with one real variable (one value per state) so an outline becomes a finding, not a locator.

Poster skeleton (fill the blanks, do not face a blank page).

- **Question:** Can a simple two-date difference map flag new surface warming?
- **Data and Methods:** Two temperature fields stored as array layers and subtracted in R; re-running the script reproduces the result.
- **Results:** A warm patch of about X degrees appeared in the northern part of the map.
- **Interpretation:** The location is consistent with a known vent, so the change is plausibly real, not noise.
- **Next Steps:** Add more dates and test whether the patch persists.

Talk talking points (a nervous beginner can say these out loud).

1. **Hook:** Volcanoes often warm at the surface before they erupt, so spotting new heat matters.
2. **Method in one breath:** I subtracted yesterday’s heat map from today’s, in a few lines of R.
3. **Most important result:** One small area in the north got hotter while everywhere else stayed the same.
4. **One honest limitation:** My data is simulated, so the obvious next step is real satellite thermal data.
5. **Next question:** How many days of warming should count as a real signal rather than random noise?

Data and reproducibility note. The demo uses the base-R `volcano` grid (no external data, no license issue). For the real project, cite the satellite thermal source (for example USGS or NASA) and keep the script, because re-running it gives the same result. That reproducibility is itself a selling point in a poster’s methods.

Next open question. Once a student can flag change between two days, the research is not finished, it has barely started: what is the smallest change this method can detect before it disappears into measurement noise? That question carries straight into Chapter 4’s work on signal over time.

F. Career Pathways

Chapter 1 teaches Earth Data Analytics, not R syntax: the durable skill is reading a sequence of numbers as a physical story (rainfall, temperature, a boundary on a map) and organizing that data so a finding can be reproduced. Those are the same first moves practitioners make on the job. The point of this section is to let a faculty member answer “where does this lead?” without inventing it.

What Chapter 1 actually builds, and where each skill is used.

Chapter 1 skill	The transferable ability	Where it is used in the real world
Vectors and <code>c()</code> , plotting two vectors	Reading a record of measurements as a picture	Hydrology (rain and stream records), air-quality monitoring, climate dashboards
Data frames, <code>head</code> , <code>str</code> , <code>names</code> , <code>\$</code>	Inspecting tabular data before trusting it	Any analyst role, environmental consulting, agency data desks
Arrays and <code>image()</code> (the two-day map)	Treating space-over-time data as stacked layers	Remote sensing, satellite monitoring, GIS analysis
Reading code inside-out	Auditing someone else’s analysis step by step	Quality control, peer review, regulatory reporting
Packages, <code>library()</code> , <code>::</code>	Extending a tool with the right add-on for the job	Every data role, because no one works in base tools alone

Chapter 1 skill	The transferable ability	Where it is used in the real world
The leaflet interactive map (optional preview)	Communicating geographic results to non-experts	Science communication, dashboards, public-facing reporting

Roles these Chapter 1 skills map to. These are entry points the chapter’s foundations genuinely support, not job guarantees. Each pairs the role with the real Earth or sustainability problem it serves.

Role / pathway	The Earth or sustainability problem it serves
Environmental data analyst	Turning monitoring records (water, air, soil) into decisions
GIS / remote-sensing technician	Reading change across maps and satellite passes over time
Hydrology or water-resources assistant	Tracking rainfall, stream flow, and groundwater records
Climate or sustainability data assistant	Following emissions and climate indicators over years and regions
Geospatial science communicator	Making maps and figures an audience can actually read
Research assistant (Earth / environmental lab)	Organizing field and sensor data so results reproduce

How to frame this for students (talking points for the instructor).

- The marketable skill is not “I know R.” It is “I can take a raw real-world record, organize it, and read what it says about the Earth.” Chapter 1 is the first rehearsal of exactly that.
- Reproducibility (re-running a script gives the same answer) is a professional standard, not a classroom nicety. Employers and journals both require it.
- The two-day change map is a miniature of real monitoring work: a before, an after, and a defensible statement about what changed.
- These skills are tool-portable. A student who learns to think this way in R carries the same thinking to Python or any agency platform; the language is the vehicle, the Earth question is the destination.

Next open question for the student. If reading two days of one volcano can hint at a career in environmental monitoring, what could you read across two decades, or across an entire country? That question is what the rest of the book answers, chapter by chapter.

G. Assess

Exam questions

Items are interpretation-first: most credit rewards reading the Earth-data result correctly, not flawless typing. All items use only Chapter 1 content (base R, `maps`, `leaflet`).

Beginner (recall and basic reading)

1. In `rainfall_mm <- c(0, 12, 5, 0, 8)`, what does the arrow `<-` do, in one sentence?
2. A point on the daily-rainfall scatter (`@fig-first-plot`) sits at (2, 12). State what day and how much rain it represents.
3. You see `[1]` at the start of an R output line. What does it tell you, and did you type it?
4. Which symbol runs a function, `( )` or `[ ]`? Which one picks a value by position?
5. True/False: a package must be re-loaded with `library()` every time you start a new R session, even though it is already installed. Explain in one line.
6. Why must a `maps` region be written as `"new jersey"` and not `"New Jersey"`?

Intermediate (apply a Chapter 1 workflow)

7. Given the data frame `df` with columns `station`, `temp`, `rain`, write the code that returns only the `temp` column, and separately the code that returns the first row.
8. Reading the volcano Day 1 / Day 2 temperature maps (`@fig-volcano-temp`): the two maps look nearly identical except for one region. Name where the temperature increased and state in one sentence what real Earth situation this could represent.
9. A classmate runs `map("state")` and gets `could not find function "map"`. State the cause and the fix.
10. Read this code inside-out and state, line by line, what each step produces:

```
result <- mean(c(10, 12, 14, 16))
result
```

11. Explain why `c(5, "number")` ends up storing `"5"` as text, and give one Earth-data example where a number should be stored as text (a label, not a quantity).

Advanced (transfer to a new but similar problem)

12. You are given two new vectors, `month <- 1:6` and `snow_cm <- c(20, 18, 9, 2, 0, 0)`. Write the code to plot them, then describe in two sentences the trend you would expect to see and what it suggests about the season.
13. You have temperature grids for three days, not two. In plain words, describe how you would store them together and how a `for` loop would let you draw all three maps without writing the `image()` line three times. (You may reference the array + loop pattern from Part C; no need to reproduce it exactly.)

14. A teammate stored station IDs as numbers and now their averages look wrong. Diagnose what went wrong using the “real burger vs picture of a burger” idea from the chapter, and state the correct way to store IDs.
15. Reading the New Jersey outline map (`@fig-nj-map`) versus the `leaflet` interactive preview (`@fig-interactive-map`): give one scientific task each is better suited to, and explain why the interactive map is labeled optional in this book.

Grading rubric

Use the **master Analysis rubric** (the project’s existing rubric for analytical work), not a new one. Chapter 1 is the Tool rung, so weight it as an early, low-stakes Analysis assessment. Do **not** invent a Chapter 1-specific rubric; apply the master Analysis rubric with the chapter weighting below. The Communication and Research master rubrics do not apply at this stage: students are not yet producing a written interpretation report (Communication) or an original question (Research). Those rubrics enter in later chapters.

Chapter 1 weighting of the master Analysis rubric (interpretation-weighted, per the book’s “Does this result make scientific sense?” goal):

Master Analysis rubric dimension	Chapter 1 weight	What earns the points here
Correct interpretation of the result	60%	Reads the rainfall pattern, the warmer volcano region, the table, or the map correctly and in plain Earth-data terms
Correct workflow / approach	25%	Right tool for the task (<code>plot</code> , <code>\$</code> , <code>[]</code> , <code>library</code> before a map); steps in a sensible order
Code runs / reproduces	15%	Code executes; minor syntax slips (a comma, a missing space) are not heavily penalized

A student who reads the science correctly but mistypes a comma still passes comfortably. A student whose code runs but who cannot say what the plot means does not. This matches the book: R is the tool, interpretation is the goal.

Assessment strategies

- **Grade the sentence, not the keystrokes.** For every plot or map item, require one plain-language sentence of interpretation (“rainfall rises then falls”; “the north of the Day 2 map is warmer”). That sentence carries most of the credit.
- **Open-book, open-R.** The chapter states students are not expected to memorize commands. Allow the book and RStudio during assessment; test reading and interpretation, not recall.

- **Two-tier partial credit.** Award the interpretation points even when the code did not run, as long as the student reasoned from a provided figure. Supply printed figures so a broken laptop never zeroes a student.
- **Quick formative checks.** Use the chapter’s built-in “Before continuing, ask yourself” and “Quick checkpoint” callouts as 2-minute exit tickets (What does the vector store? How many values? Which is at position 1?).
- **Keep leaflet and Part C out of graded recall.** Assess them, if at all, only as “explain the idea” transfer items, never as syntax to reproduce.
- **Reuse the chapter Exercises (1-5) as the baseline homework,** then add one transfer item (like exam item 12) so students apply the workflow to data they have not seen.

H. Adapt and support

Level modifications (high school / undergraduate / graduate)

Aspect	High school	Undergraduate	Graduate
Scope	Part A only + one maps outline	Part A + Part B (maps, packages)	Parts A-C including arrays, loops, volcano application
Pace	Two sessions; lots of run-it-together	One to two sessions	One session; assign Part C as pre-reading
Emphasis	“We can store data and see it as a picture”	“R is a reproducible tool for Earth data”	“Why arrays and loops scale to many time steps and grids”
Assessment	Beginner items only; interpretation in plain words	Beginner + intermediate items	Add advanced transfer items (12-15)
Earth framing	Rainfall and a map of their own state	Rainfall, station table, US/Greece maps	Volcano Day 1 / Day 2 change as a model for time-varying Earth data
leaflet	Demo only, “look what is possible”	Demo, label optional	Optional self-study; tie to dashboards and science communication

In all three levels, the subject is the Earth-data idea; R is the means. Do not escalate “rigor” by adding programming theory (loops, control flow for their own sake). Escalate by adding harder interpretation and more transfer to new datasets.

Online teaching adaptations

- **Pre-record the three key runs** (rainfall scatter, station data frame, one **maps** outline) as 3 short clips. The chapter has no live downloads, so recordings will always match what students

get.

- **Post the pre-rendered figures** (@fig-first-plot, @fig-us-map/@fig-nj-map, @fig-volcano-temp) in the LMS so asynchronous students interpret them even before they install R.
- **Share a starter script** containing the chapter's code blocks in order, so students run top-to-bottom and hit the steps in sequence (prevents the "object not found" error).
- **Use the callouts as discussion prompts.** "Longitude first, latitude second" and "Common beginner mistakes" make good async forum questions.
- **Office-hours triage sheet** = the troubleshooting table above; paste it as a pinned post so most errors self-resolve.

Flipped-classroom adaptations

- **Before class (at home):** students install R + RStudio, watch the 3 recorded runs, and read Part A. Pre-class check: submit one screenshot of their own rainfall scatter.
- **In class (active):** predict-then-run. Show `month/snow_cm` (exam item 12), have students predict the trend on paper, then run it and compare. Build the data frame together and ask which column answers a given Earth question.
- **After class:** chapter Exercises 1-5 plus one transfer item. Part C (lists, arrays, loops, volcano) becomes optional enrichment for students who want more.
- **Confidence move for the instructor:** because everything is pre-recorded and there are no downloads, you can teach the in-class portion entirely from the pre-rendered figures if any machine misbehaves, and never lose the thread.

Student-support strategies

- **Normalize errors early.** The chapter says "Errors are normal when learning programming." Repeat it. Show the lone-+-then-Esc fix live so students see recovery, not failure.
- **Give the plain-words anchor for every term** the first time it appears (box for object, short list for vector, position for index, tool for function). Never define a term with another undefined term.
- **WHY before HOW.** Always state the Earth-data reason ("we want to see the pattern as a picture") before showing the `plot()` line. A student who knows why will tolerate unfamiliar syntax.
- **Provide the starter script and a one-page command map** (`<-`, `c()`, `[ ]`, `()`, `$`, `head`, `str`, `library`) so no one is blocked by recall.
- **Reassure about memorization.** Point students to the chapter's repeated "you are not expected to memorize" callouts; grade reading and interpretation, which lowers anxiety.
- **Pair-programming and tab-completion.** Teach the Tab autocomplete tip (for example `df$` then Tab) early; it prevents typos and the resulting cryptic errors, which are the top confidence-killer for beginners.
- **A safe "I do not know" script for the instructor:** "Good question, let us look it up with `?functionname` together." Modeling the help page turns a confidence risk into a teachable habit.